

Locking Away Prosperity? Evaluating the Economic Impact of the Key to NYC Program

1 Introduction

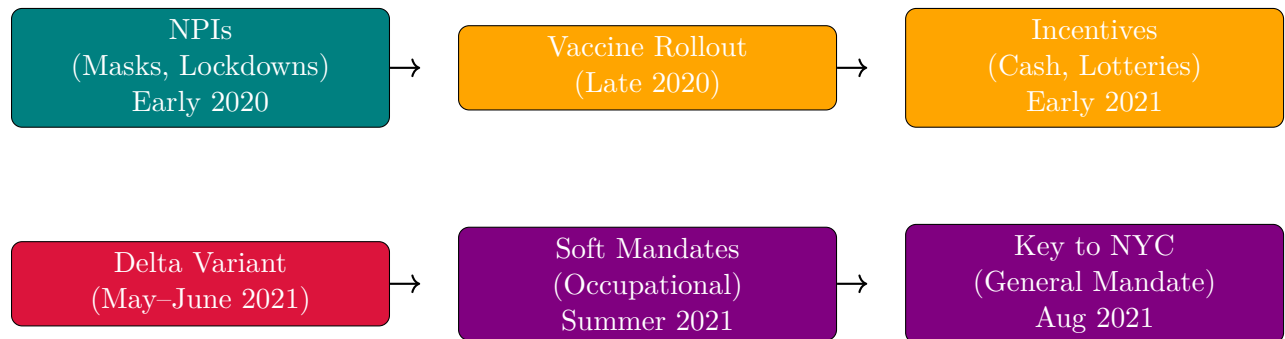
The COVID-19 pandemic triggered a global wave of non-pharmaceutical interventions (NPIs) as governments acted swiftly to limit viral transmission. Much early research focused on the public health rationale for these policies—especially the importance of early action to contain exponential spread. For instance, [Friedson et al. \[2021\]](#) and [Yang \[2021\]](#) use synthetic control methods to evaluate lockdowns in California and Wuhan, emphasizing the critical role of timing. Similarly, [Bayat et al. \[2020\]](#) estimate that New York City could have reduced COVID-19 deaths by over 80% had its first lockdown occurred just ten days earlier. Subsequent work examined the economic consequences of such interventions. While NPIs helped flatten the curve, they also raised concerns about business closures, layoffs, and lasting economic scarring. Studies like [Gibson and Sun \[2023\]](#) and [Lee and Yang \[2022\]](#) document labor market costs of lockdowns, while [Ke and Hsiao \[2022\]](#) show that although strict NPIs incur short-term economic losses, those effects may be temporary. Vaccine mandates emerged as a less restrictive (but still controversial) policy tool during later phases of the pandemic. Unlike lockdowns, mandates aimed to curb transmission without shutting down businesses. Nevertheless, critics argued that mandates could hamper fragile economic recoveries by alienating customers or exacerbating staffing shortages in consumer-facing sectors like restaurants. Supporters countered that mandates would improve public safety and restore consumer confidence. These competing narratives came to a head with New York City’s *Key to NYC* program—implemented in August 2021—the first large-scale vaccine mandate for indoor commercial activity in the U.S. [[Towey, 2021](#), [Rogers, 2021](#), [Bardosh et al., 2022](#)].

This paper evaluates the causal effect of the Key to NYC mandate on employment in New York City’s full-service restaurant sector—the segment most directly affected by the policy. While prior research has analyzed mandates’ effects on vaccination rates and health outcomes [[Cohn et al., 2022](#), [Acharya and Dhakal, 2021](#), [Lang et al., 2023](#)], relatively little is known about their labor market consequences, particularly in industries that rely heavily on face-to-face interaction. This study helps fill that gap. Because randomized experiments are infeasible, this paper adopts a synthetic control approach to estimate the counterfactual trajectory of employment absent the mandate. Using MSA-level monthly data from the Federal Reserve Bank of St. Louis, I construct a synthetic control for NYC’s year-over-year

growth in full-service restaurant employment. The donor pool consists of 29 MSAs that did not implement similar mandates during the same period.¹

The rest of the paper is organized as follows. Section 2 provides background on the evolution of pandemic interventions and the Key to NYC program. Section 3 describes the employment data and outcome construction. Section 4 outlines the empirical strategy, including the synthetic control methodology. Section 5 presents the main findings and robustness checks. Section 6 interprets the results and discusses limitations. Section 7 concludes.

2 Policy Background



Timeline of major pandemic policy phases

2.1 NPIs Before Vaccines

NPIs had aimed to reduce transmission by limiting contact or promoting respiratory barriers. The first and most famous of these interventions was the 2020 lockdown in Wuhan, China, where local officials went as far as to forcibly seal people in their homes to reduce the possibility to spread the virus; similar actions were taken in Europe and India. Even in the United States, states that could effectively seal their borders, such as Hawaii, did so, with Hawaii mandating two-week quarantines to visitors, which effectively amounted to a tourism ban. Mask mandates were widely adopted as well due to their low cost and direct mechanism of action [Hansen and Mano, 2023, Chernozhukov et al., 2021, Mitze et al., 2020]. While not as severe as lockdowns, the idea of mask mandates was to allow for some freedom of movement while minimizing the chance to spread the virus further. While both sets of policies were effective, they provoked political and economic backlash due to their disruptive nature.

2.2 The First Vaccine Policies

The arrival of vaccines in late 2020 ushered in a new phase of pandemic governance, giving jurisdictions a wider array of policy options to employ. Now, instead of lockdowns, governments finally had pharmaceutical interventions in their toolkit. For the first few months, it worked: when vaccines were released, initial uptake was high, particularly among older

¹For example, 'SMU11000007072251101SA' (Washington, DC) or 'SMU08197407072251101SA' (Denver MSA).

adults and (often mandated for) healthcare workers. However, as eligibility expanded in early 2021, demand slowed amid misinformation, mistrust, and logistical barriers. In response, policymakers deployed incentive-based strategies to encourage vaccination [Barber and West, 2022, Robertson et al., 2021]. These included direct cash payments, vaccine lotteries (e.g., Ohio’s “Vax-a-Million” from May to June [Fuller et al., 2022]), and smaller symbolic giveaways. One paper dubbed this the era of “donuts, drugs, booze, and guns,” reflecting the extraordinary lengths to which governments went to boost uptake through rewards rather than requirements [Tinari and Riva, 2021].

Despite their creativity, most incentives had modest or short-lived effects [Saban et al., 2021, Walkey et al., 2021, Khazanov et al., 2023]. Many were poorly targeted, administratively complex, or insufficient to overcome deeper resistance. The emergence of the highly transmissible Delta variant in mid-2021 raised the stakes. Initially identified in India, Delta reached the United States by May and accounted for the majority of new cases by June. Unlike in 2020, officials now had vaccines at scale—but not enough vaccinated people. Given a new more contagious strain, soft mandates emerged, targeting workers in high-risk occupations such as healthcare and education. These policies were not general mandates but did make vaccination a condition of continued employment [American Medical Association et al., 2021]. Over the summer, local governments and organizations expanded this approach [American Medical Association et al., 2021, Al Jazeera, 2021]. By August 2021, New York City introduced the Key to NYC program—one of the first large-scale vaccine mandates in the U.S. applied to the general public. Unlike prior efforts that relied on encouragement, Key to NYC made access to indoor dining, fitness, and entertainment conditional on vaccination status. It represented a new policy mechanism: exclusion rather than incentive.

2.3 The First Vaccine Mandates

Per the above, vaccine mandates could influence employment in full-service restaurants through both supply- and demand-side channels, with theoretically ambiguous net effects. On the supply side, some workers may exit the labor force or shift sectors if unwilling to comply, especially in lower-wage service jobs where vaccine hesitancy may be higher. However, others may comply with the mandate precisely *because* they wish to keep their jobs, choosing continued employment over personal objection. That is, the employment effect may hinge less on individual preferences in isolation and more on the strength of the economic incentives at stake. Employers, meanwhile, may face frictional costs from enforcing mandates or accommodating staffing turnover. On the demand side, mandates may increase consumer confidence by signaling a safer environment, thereby encouraging more patrons to return to indoor dining. For a sector like full-service restaurants, where both interpersonal contact and discretionary spending are central, mandates could plausibly stimulate demand even as they introduce short-run supply constraints. Whether employment rises or falls in net depends on which margin dominates. In short, this analysis seeks to quantify the net effect of the mandate on restaurant employment, capturing the balance between these potentially opposing forces.

3 Dataset

We use monthly MSA-level employment series from the Federal Reserve Bank of St. Louis (FRED) and, by extension, the Bureau of Labor Statistics, spanning January 1991 through December 2022, yielding $T = 384$ monthly observations per unit. Our analysis focuses on three related outcome sets: (1) full-service restaurant employment for 30 MSAs including New York City as the treated unit with 29 donor MSAs; (2) broader leisure industry employment at the MSA level for approximately 255 MSAs from 1991 to December 2022. For the broader leisure employment analysis, New York City (FRED code SMU36935617072251101SA) is the treated unit, with the remaining MSAs as the donor pool. The broader leisure industry MSA analysis extends this framework to approximately 255 MSAs observed through December 2022 (FRED code SMU36000007072200001SA), providing a larger donor pool and outcome set.

Let e_{jt} denote employment in unit j at month t . For both the full-service restaurant analysis and broader leisure market analysis, the outcome is constructed as the year-over-year growth rate in percentage points,

$$y_{jt} = 100 \cdot \frac{e_{jt} - e_{j,t-12}}{e_{j,t-12}}, \quad j = 1, \dots, N, \quad t = 13, \dots, T.$$

of the smoothed and seasonally adjusted values. Outcomes are collected into vectors $\mathbf{y}_j = (y_{j1}, \dots, y_{jT})^\top$, and the donor pool is defined as $\mathcal{N}_0 = \mathcal{N} \setminus \{1\}$ with $N_0 = 29$. The *Key to NYC* policy, implemented in August 2021, designates July 2021 as the last pre-treatment month T_0 , with the treatment indicator defined by

$$d_{jt} = \mathbf{1}\{j = 1 \wedge t > T_0\}.$$

Observed outcomes satisfy

$$y_{jt} = (1 - d_{jt})y_{jt}(0) + d_{jt}y_{jt}(1),$$

with causal effects $\tau_{jt} = y_{jt}(1) - y_{jt}(0)$ and post-treatment average treatment effect on the treated (ATT)

$$\text{ATT} = \frac{1}{|\mathcal{T}_2|} \sum_{t \in \mathcal{T}_2} \tau_{1t},$$

where $\mathcal{T}_2 = \{t > T_0\}$.

4 Econometric Methodology

To estimate the counterfactual employment trajectory for NYC's full-service restaurant sector and the broader employment trends in the leisure labor market, I employ a synthetic control

approach that hybridizes Forward Selection SCM (FSCM) and the Augmented SCM (ASCM) estimators. The classical Synthetic Control Method (SCM) solves

$$\mathbf{w}^{\text{SCM}} = \underset{\mathbf{w} \in \mathcal{W}_{\text{conv}}}{\text{argmin}} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \mathbf{w} \right\|_2^2,$$

where

$$\mathcal{W}_{\text{conv}} = \left\{ \mathbf{w} \in \mathbb{R}_{\geq 0}^{N_0} \mid \mathbf{1}^\top \mathbf{w} = 1 \right\}$$

is the probability simplex. This finds the weights $\mathbf{w}^{\text{SCM}}$ that best reconstruct the treated unit's pre-treatment outcomes as a convex combination of donor outcomes.

4.1 Forward Selection SCM

FSCM constructs the synthetic control iteratively by adding donor units one at a time. Starting with an empty selected donor set $\mathcal{S} = \emptyset$, at each iteration the algorithm evaluates each candidate donor $j \in \mathcal{N}_0 \setminus \mathcal{S}$ by forming an expanded set $\mathcal{S}' = \mathcal{S} \cup \{j\}$. For each $\mathcal{S}'$, it solves the restricted SCM problem over the donor submatrix $\mathbf{Y}_0^{\mathcal{S}'}$, consisting of columns corresponding to units in $\mathcal{S}'$:

$$\mathbf{w}_{\mathcal{S}'}^* = \underset{\mathbf{w} \in \mathcal{W}_{\text{conv}}(\mathcal{S}')}{\text{argmin}} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1, \mathcal{S}'} \mathbf{w} \right\|_2^2,$$

where

$$\mathcal{W}_{\text{conv}}(\mathcal{S}') = \left\{ \mathbf{w} \in \mathbb{R}_{\geq 0}^{|\mathcal{S}'|} \mid \mathbf{1}^\top \mathbf{w} = 1 \right\}.$$

The donor selected at iteration k is

$$j^* = \underset{j \in \mathcal{N}_0 \setminus \mathcal{S}}{\text{argmin}} \min_{\mathbf{w} \in \mathcal{W}_{\text{conv}}(\mathcal{S} \cup \{j\})} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1, \mathcal{S} \cup \{j\}} \mathbf{w} \right\|_2^2,$$

and $\mathcal{S}$ is updated as

$$\mathcal{S} \leftarrow \mathcal{S} \cup \{j^*\}.$$

This forward selection continues until a stopping criterion is met, such as reaching a maximum number of donors, a minimum improvement threshold, or an information criterion like modified BIC.

4.2 The Augmented Synthetic Control Estimator

Building on the baseline SCM, ASCM relaxes the convex weight restriction by allowing affine weights and adds a regularization penalty that shrinks the solution towards a reference weight vector $\mathbf{w}_0$, typically the SCM or FSCM solution. The augmented objective is

$$\mathbf{w}^{\text{aug}} = \underset{\mathbf{w} \in \mathcal{W}_{\text{aff}}}{\operatorname{argmin}} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \mathbf{w} \right\|_2^2 + \lambda \left\| \mathbf{w} - \mathbf{w}_0 \right\|_2^2,$$

where $\lambda \geq 0$ controls the penalty strength and

$$\mathcal{W}_{\text{aff}} = \left\{ \mathbf{w} \in \mathbb{R}^{N_0} \mid \mathbf{1}^\top \mathbf{w} = 1 \right\}$$

is the affine subspace of weights summing to one, but not restricted to be non-negative. The intuition is that when $\mathbf{w}_0$ (e.g., FSCM weights) provides a good fit, λ will be large to maintain interpretability; when fit needs improvement, λ can be small to allow extrapolation beyond the convex hull.

4.3 Forward Augmented Synthetic Control

FASC integrates the interpretability of FSCM with the flexibility of ASCM by first performing forward selection to obtain $\mathbf{w}^{\text{FSCM}}$, then solving

$$\mathbf{w}^{\text{FASC}} = \underset{\mathbf{w} \in \mathcal{W}_{\text{aff}}}{\operatorname{argmin}} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \mathbf{w} \right\|_2^2 + \lambda \left\| \mathbf{w} - \mathbf{w}^{\text{FSCM}} \right\|_2^2,$$

where $\lambda \geq 0$ balances fidelity to the sparse forward-selected weights and pre-treatment fit.

Large λ yields a solution close to $\mathbf{w}^{\text{FSCM}}$, preserving sparsity and interpretability, whereas small λ prioritizes fit, allowing positive or negative weights outside the original selected donors.

4.3.1 Cross-Validation for λ

The hyperparameter λ is chosen by time-split cross-validation over the pre-treatment period. The training set (first half of pre-treatment) is used to estimate

$$\mathbf{w}^{\text{FASC}}(\lambda) = \underset{\mathbf{w} \in \mathcal{W}_{\text{aff}}}{\operatorname{argmin}} \left\| \mathbf{y}_{\text{train}} - \mathbf{Y}_{0,\text{train}} \mathbf{w} \right\|_2^2 + \lambda \left\| \mathbf{w} - \mathbf{w}^{\text{FSCM}} \right\|_2^2,$$

while the validation set (second half of pre-treatment) evaluates out-of-sample error via

$$\text{CV-RMSE}(\lambda) = \left\| \mathbf{y}_{\text{val}} - \mathbf{Y}_{0,\text{val}} \mathbf{w}^{\text{FASC}}(\lambda) \right\|_2.$$

The λ minimizing CV-RMSE balances interpretability and fit. Bayesian optimization over $\log_{10}(\lambda) \in [-2, 3]$ with Gaussian Process surrogates and Expected Improvement acquisition

efficiently explores this space. Since $\mathbf{w}^{\text{FSCM}}$ is feasible in the FASC optimization, the in-sample Mean Squared Error (MSE) satisfies

$$\text{MSE}^{\text{FASC}} = \frac{1}{T_0} \|\mathbf{y}_1^{T_1} - \mathbf{Y}_0^{T_1} \mathbf{w}^{\text{FASC}}\|_2^2 \leq \frac{1}{T_0} \|\mathbf{y}_1^{T_1} - \mathbf{Y}_0^{T_1} \mathbf{w}^{\text{FSCM}}\|_2^2 = \text{MSE}^{\text{FSCM}},$$

ensuring FASC’s fit is never worse and typically better than FSCM.

5 Results

6 Discussion

7 Conclusion

References

- Binod Acharya and Chandra Dhakal. Implementation of state vaccine incentive lottery programs and uptake of covid-19 vaccinations in the united states. *JAMA Network Open*, 4(12):e2138238–e2138238, 12 2021. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2021.38238.
- Al Jazeera. Covid-19 delta surge in us leads to new restrictions, jab push, July 2021. URL <https://tinyurl.com/yhbl97yx>. Officials are growing more worried about the current rise in infections and are blaming the unvaccinated for the spike.
- American Medical Association et al. Major health care professional organizations call for covid-19 vaccine mandates for all health workers. <https://tinyurl.com/24r6jjtv>, July 2021. Press Release, July 26, 2021.
- Andrew Barber and Jeremy West. Conditional cash lotteries increase covid-19 vaccination rates. *Journal of health economics*, 81:102578, 2022.
- Kevin Bardosh, Alex de Figueiredo, Rachel Gur-Arie, Euzebiusz Jamrozik, James Doidge, Trudo Lemmens, Salmaan Keshavjee, Janice E. Graham, and Stefan Baral. The unintended consequences of covid-19 vaccine policy: why mandates, passports and restrictions may cause more harm than good. *BMJ Global Health*, 7(5):e008684, 2022. doi: 10.1136/bmjgh-2022-008684. URL <https://doi.org/10.1136/bmjgh-2022-008684>.
- Niloofar Bayat, Cody Morrin, Yuheng Wang, and Vishal Misra. Synthetic control, synthetic interventions, and covid-19 spread: Exploring the impact of lockdown measures and herd immunity, 2020. URL <https://arxiv.org/abs/2009.09987>.
- Victor Chernozhukov, Hiroyuki Kasahara, and Paul Schrimpf. Causal impact of masks, policies, behavior on early covid-19 pandemic in the u.s. *Journal of Econometrics*, 220(1): 23–62, 2021. doi: 10.1016/j.jeconom.2020.09.003.

- Ezra Cohn, Michael Chimowitz, Theodore Long, Jay K Varma, and Dave A Chokshi. The effect of a proof-of-vaccination requirement, incentive payments, and employer-based mandates on covid-19 vaccination rates in new york city: a synthetic-control analysis. *The Lancet Public Health*, 7(9):e754–e762, 2022.
- Andrew I. Friedson, Drew McNichols, Joseph J. Sabia, and Dhaval Dave. Shelter-in-place orders and public health: Evidence from california during the covid-19 pandemic. *Journal of Policy Analysis and Management*, 40(1):258–283, 2021. doi: 10.1002/pam.22267.
- Sam Fuller, Sara Kazemian, Carlos Algara, and Daniel J. Simmons. Assessing the effectiveness of covid-19 vaccine lotteries: A cross-state synthetic control methods approach. *PLOS ONE*, 17(9):1–18, 09 2022. doi: 10.1371/journal.pone.0274374.
- John Gibson and Xiaojin Sun. A synthetic control analysis of us state level covid-19 stay-at-home orders on new jobless claims. *Journal of Economics and Finance*, 47(1):1–14, 2023.
- Niels-Jakob H. Hansen and Rui C. Mano. Mask mandates save lives. *Journal of Health Economics*, 88:102721, 2023. ISSN 0167-6296. doi: <https://doi.org/10.1016/j.jhealeco.2022.102721>. URL <https://www.sciencedirect.com/science/article/pii/S0167629622001357>.
- Xiao Ke and Cheng Hsiao. Economic impact of the most drastic lockdown during covid-19 pandemic—the experience of hubei, china. *Journal of Applied Econometrics*, 37(1):187–209, 2022. doi: 10.1002/jae.2871.
- Gabriela K. Khazanov, Rebecca Stewart, Matteo F. Pieri, Candice Huang, Christopher T. Robertson, K. Aleks Schaefer, Hansoo Ko, and Jessica Fishman. The effectiveness of financial incentives for covid-19 vaccination: A systematic review. *Preventive Medicine*, 172:107538, 2023. doi: 10.1016/j.ypmed.2023.107538.
- David Lang, Lief Esbenshade, and Robb Willer. Did ohio’s vaccine lottery increase vaccination rates? a pre-registered, synthetic control study. *Journal of Experimental Political Science*, 10(2):242–260, 2023. doi: 10.1017/XPS.2021.32.
- Jongkwan Lee and Hee-Seung Yang. Pandemic and employment: evidence from covid-19 in south korea. *Journal of Asian economics*, 78:101432, 2022.
- Timo Mitze, Reinhold Kosfeld, Johannes Rode, and Klaus Wälde. Face masks considerably reduce covid-19 cases in germany. *Proceedings of the National Academy of Sciences*, 117(51):32294–32301, 2020. doi: 10.1073/pnas.2015954117.
- Christopher Robertson, K Aleks Schaefer, and Daniel Scheitrum. Are vaccine lotteries worth the money? *Economics Letters*, 209:110097, 2021.
- Kate Rogers. Restaurants grapple with vaccine mandate policies in historically tight labor market, 2021. URL <https://tinyurl.com/yemwnsgr>. Published by CNBC, August 27, 2021. Accessed 2025-08-20.
- Mor Saban, Vicki Myers, Shani Ben Shetrit, and Rachel Wilf-Miron. Issues surrounding incentives and penalties for covid-19 vaccination: The israeli experience. *Preventive*

- Medicine*, 153:106763, 2021. ISSN 0091-7435. doi: <https://doi.org/10.1016/j.ypmed.2021.106763>. URL <https://www.sciencedirect.com/science/article/pii/S0091743521003327>.
- Serena Tinari and Catherine Riva. Donuts, drugs, booze, and guns: what governments are offering people to take covid-19 vaccines. *BMJ*, 374, 2021. doi: 10.1136/bmj.n1737. URL <https://www.bmj.com/content/374/bmj.n1737>.
- Robert Towey. New york city mayor bill de Blasio will mandate covid vaccines for certain indoor activities, 2021. URL <https://tinyurl.com/yh8hasxz>. Published by CNBC, August 3, 2021. Accessed 2025-08-20.
- Allan J. Walkey, Anica Law, and Nicholas A. Bosch. Lottery-based incentive in ohio and covid-19 vaccination rates. *JAMA*, 326(8):766–767, 08 2021. doi: 10.1001/jama.2021.11048.
- Xiaoxuan Yang. Does city lockdown prevent the spread of covid-19? new evidence from the synthetic control method. *Global Health Research and Policy*, 6:1–14, 2021.